

Testing and Debugging

Manual tests, debug prompts, PR template, release checks

Per-Phase Testing Checklist

After every phase, check:

1. Godot opens the project.
2. Main.tscn loads.
3. Press Play.
4. No script parse errors.
5. No missing scene reference errors.
6. Player still appears.
7. Player still moves.
8. Player can still shoot if shooting is implemented.
9. Enemies still spawn if spawner is implemented.
10. Bullets still collide correctly.
11. Score/lives/HUD still update.
12. Pause/game over still works if implemented.
13. No console spam.
14. No huge performance drop.
15. docs/known_issues.md is updated.

Debugging Prompts

When Godot throws an error, copy the exact error into one of these prompts. Keep fixes narrow and targeted.

General Error

The Godot project has this error:

[paste exact error]

Please fix the smallest possible cause.

Do not rewrite unrelated systems.

Explain which file/path caused it.

Keep the project runnable.

Update docs/known_issues.md if the issue reveals an unfinished system.

Scene Reference Error

Godot reports a missing or invalid scene/script reference:

[paste exact error]

Inspect the related .tscn and .gd files.

Fix resource paths, node paths, exported scene references, or preload paths as needed.

Do not change gameplay behaviour unless required.

Collision Not Working

Collision is not working correctly:

Observed behaviour:

[describe what happens]

Expected behaviour:

[describe expected result]

Check:

- Area3D setup
- CollisionShape3D
- collision layers/masks
- signal connections
- bullet hit logic
- enemy take_damage()

- player take_damage()

Fix the minimum necessary code and document the cause.

Player Movement Broken

Player movement is broken:

Observed:
[describe]

Expected:
The player should move on the X/Z plane using move_left, move_right, move_up, move_down and stay inside playfield bounds.

Check player.gd, input actions, Main.tscn node references, and project.godot input map.
Fix only the movement/input issue.

Json Wave Loading Broken

Enemy wave loading is broken:

Observed:
[describe]

Expected:
EnemySpawner should load data/waves/stage_01_waves.json and spawn enemies at the specified times.

Check:
- JSON file path
- JSON syntax
- FileAccess usage
- parsed data validation
- enemy type mapping
- scene preload/instantiate logic

Fix the loader and add graceful error messages.

Performance Issue

Performance drops when many bullets or effects are active.

Please inspect:
- bullet creation/destruction
- enemy bullet creation/destruction
- effect spawning
- cleanup logic
- object pooling

Implement the smallest safe optimization.
Do not change gameplay feel.

Pull Request Template

Summary

Describe what this PR changes.

Phase

Which development phase does this PR implement?

Gameplay Impact

- [] No gameplay change
- [] Player movement
- [] Shooting
- [] Enemies
- [] Collision
- [] HUD
- [] Stage/waves
- [] Effects/audio
- [] Menus/game states
- [] Performance/export

Test Checklist

- [] Project opens in Godot
- [] Main scene runs
- [] No script parse errors
- [] No missing critical scene references
- [] Feature works in a basic manual test
- [] Existing features still work
- [] README/docs updated if needed

Known Issues

List known unfinished items or bugs.

Screenshots / Notes

Add screenshots or gameplay notes if relevant.

Release Checklist

- Project opens in the target Godot version.
- Main scene runs without parse errors.
- One full stage is playable from title to mission complete.
- Game over and restart work.
- Score, lives, bombs, and stage UI update correctly.
- Bullets, effects, and enemies clean up or return to pools.
- Placeholder assets are clearly marked.
- No copied copyrighted assets are included.
- Export guide and testing checklist are updated.